May 25, 2026

Triple plaque purification to generate clonal phage lysates

DOI
https://dx.doi.org/10.17504/protocols.io.4r3l2b63l1y9/v1

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Protocol Citation: Melissa Duhaime 2026. Triple plaque purification to generate clonal phage lysates. **protocols.io**
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Protocol status: Working
We use this protocol and it's working

Created: October 17, 2017

Last Modified: May 25, 2026

Protocol Integer ID: 8291

Abstract

Use to purify plaques to attain a lysate containing a clonal population of phages.

Purification Rounds 1-3

1 Start with an overnight agar overlay plate that shows plaque formation.

Using a 200 ul pipette tip, pick a plaque from the original host screening plate that plaques were discovered on. Choose a plaque that is similar in morphology and size to the majority of the other plaques on the plate.

Mark on the plate which plaque was picked for future reference.

2 Vortex the pipette tip in 1 mL of phage buffer. Let the sample stand for at least 30 minutes. Vortex again before plating sample.

00:30:00 bringing phage into solution

3 Plate dilutions of 10^{-2} - 10^{-6}. This will help to ensure countable data after incubation.

Incubate overnight at room temperature.

16:00:00 overnight incubation (can be 16-28 hours)

4 Pick a plaque from overnight plates for Purification Round 2.

5 Repeat steps above for one more round of purification. This is purification 3.

16:00:00 overnight incubation (can be 16-28 hours)

Generate clonal lysate

6 Pick plaque from Purification 3 plate using a 200 µl pipette tip. Vortex the pipette tip in 1 mL of phage buffer. Let the sample stand for at least 30 minutes. Vortex again before plating sample.

00:30:00 bringing phage into solution

7 Using a dilution that will lyse 95% of the host lawn, incubate for 24 hours at RT.

24:00:00 overnight incubation

Collect Lysate

8 Identify plates that are 95% lysed. They can be as low as 70%, but completely lysed plates should not be used. Hold plate at an angle, so as not to disturb top agar. Gently

pipette 5 mL of phage buffer onto plate. Place on platform shaker at 70 rpm for 30 min to 3 hr at room temperature.

00:30:00 can be 30 mins to 3 hrs

9 Phages should now be suspended in buffer. Use a serological pipette to extract liquid into a 15 mL conical tube. If there is solid matter in the sample, centrifuge the sample at 3220 g for 5 min and extract the supernatant.

10 Using a 5-10 mL syringe, filter the sample into a new 15 mL conical tube using a 0.8/0.2 um filter (blue and green).

Determine titer of lysate

11 Plate plaque assays on corresponding hosts, at dilutions 10^{-1} - 10^{-9}.

Incubate plate for 24 hours.

24:00:00 incubation

12 Count plaques the following day. If there are fewer than 3 × 10-9 phage / mL, make multiple plate lysates of that phage and concentrate the combined lysate with PEG or Amicon filtration cartridges.